Practice Improves Performance of a 2D Uncertainty Integration Task Within and Across Visualizations

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Abstract—Information uncertainty is ubiquitous in everyday life, including in domains as diverse as weather forecasts, investments, and health risks. Knowing how to interpret and integrate this uncertain information is vital for making good decisions, but this can be difficult for experts and novices alike. In this study, we examine whether brief, focused practice can improve people's ability to understand and integrate bivariate Gaussian uncertainty visualized via ensemble displays, summary displays, and distributional displays, and we examine whether this is influenced by the complexity of the displayed information. In two experiments (N=118 and 56), decision making was faster and more accurate after practice relative to before practice. Furthermore, the performance improvements transferred to use of display types that were not practiced. This suggests that practice with feedback may improve underlying skills in probabilistic reasoning and provides a promising approach to improve people's decision making under uncertainty.

Index Terms—Decision making, training, visualization, uncertainty

1 Introduction

Making decisions under uncertainty is a difficult but pervasive task, occurring in contexts as diverse as financial decisions, disaster response, and medical decision-making. Problems with audience comprehension of data uncertainty have led many communicators to simply forego including uncertainty estimates at all [1]. This approach can lead to maladaptive consequences when decision makers do not have access to the uncertainty information they need, ranging from wasting time waiting for a bus [2] to making decisions that can lead a country to war [3, box 1]. One way of presenting uncertain information is by utilizing visualizations. An example of a common use for visualizations in presenting uncertainty information can be found in conveying geographic information, such as approximating the location of a device through GPS. Visualizations are also used to present various types of uncertainty information about weather, betting odds, or financial risk.

Various types of training have been explored to help users understand data visualizations and integrate uncertainty to make decisions. These include approaches such as the provision of verbal instructions, the use of animations, allowing interactivity with visualizations, and encouraging thought experiments and re-creations of visualizations [4], [5], [6], [7], [8], and these training methods have demonstrated various degrees of effectiveness in enhancing understanding of uncertainty information. Additionally, increased understanding in one type of visualization can transfer to other familiar types of visualization [9]. Making decisions based on visualized uncertainty is a two-step process and depends on extracting information from the graphic and then using that information to make some inference [10]. Identifying a brief and effective method to train both of these skills (the extraction of uncertainty information from visualizations, plus the reasoned inference based on the extracted information) could open the door to better decision making across a range of real-world contexts.

To identify effective means of training, we examined the use of practice with feedback and explored two main hypotheses: 1) Having users practice a decision-making task with uncertainty visualizations, followed by prompt performance feedback, should enhance performance with that same type of visualization, as would be demonstrated in pre- vs. post-test performance; and 2) Deliberate practice with a probabilistic reasoning task should improve performance of the underlying inference, as would be demonstrated by improved transfer performance on different (untrained) visualization types. Our experimental task is a simplified and abstracted representation of a real-world situation in which a decision-maker needs to consider multiple uncertain

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reports when judging the most likely or best location for an action. This kind of situation arises in search and rescue, wayfinding, law enforcement, and military operations.

We utilized three different ways of presenting uncertainty information to participants:

1) Ensemble displays, which present some number of cases as representative of the underlying distribution, such as showing some number of random draws from the distribution. Examples include representative subset samples [11], quantile dotplots [2], and representative track ensembles [12]. Such ensemble displays are thought to be beneficial for users because the majority, if not all, of the data is depicted [13]. However, it might lead a user to put disproportionate weight on the realized sample, rather than considering it just one of many possible samples [14].
2) Summary displays, which are displays that show statistical summaries of their underlying distributions, e.g., error bars or confidence intervals. Summary displays are concise, in that they convey distributional information with a minimal (or nearly so) amount of graphical information. However, summary displays often have an effect of creating apparent boundaries in spaces that are better understood as continuous. Displaying continuous space as partitioned can invite errors of reasoning [15], [16].
3) Distributional displays, which display the entirety of a probability distribution. For example, using a curve to display a one-dimensional probability density function. Other approaches use color saturation, value, or opacity to represent probability density. These color-based approaches can be used with one- and two-dimensional probability distributions. However, human color perception does not map straightforwardly to color spaces that are convenient for data display and there are limitations with color perception and sensitivity as it is displayed in print or screens [17].

Given that we utilized three visually distinct ways to represent uncertainty data, our study consists of two experiments: in the first experiment, we tested whether deliberate practice with an ensemble representation would improve performance with that representation and the others. Ensemble representations are considered the most intuitive, so we reasoned that this was the most likely to be accessible and lead to the best chance of improving the underlying inference. In the second experiment, we tested a summary representation, reasoning that as the most abstract representation, it would be most likely to demonstrate type-specific effects.

2 Evaluation Method

Here we describe the overarching method that underlies two separate studies. Although both studies follow the same experimental design, differences will be detailed in the following sections (see Experiment 1, Experiment 2). Data collection method, sample size, hypotheses, and analysis approach for these experiments were pre-registered [https://osf.io/yqun6/ and https://osf.io/x5u4b/], however the analysis plan did not account for differences in stimulus difficulty, which turned out to be important. The analyses presented here differ substantially from the registered analysis plan.

2.1 Procedure

We used the service Prolific Academic to advertise the web-based experiment to eligible prospective participants based on self-report responses regarding normal vision, normal color vision, normal hearing, fluency in English, and age 18 or older. After enrollment, participants were directed to a Qualtrics survey to confirm eligibility, provide informed consent, respond to demographic questions, and complete a 14-plate Ishihara color vision test. A technology questionnaire was administered to ensure participants were completing the experiment on a desktop computer or laptop. These criteria were used to ensure all participants could equally perceive all aspects of the experiment. Next, participants were prompted to resize an image on their screen to match the size of a standard physical card (3.375 in x 2.125 in, such as a credit card) to calibrate task stimuli size [18]. After vision screening, the Regulatory Focus Questionnaire [19], Big Five Inventory [20], [21], Game Use Inventory [22], Graph Literacy [23], and Numeracy [24] questionnaires were administered. After the questionnaires, participants watched an 8-minute instruction video that described three visualization types, how to interpret areas with high and low uncertainty, and the task objective–to find the most likely hidden location.

Next, participants answered 13 multiple choice comprehension questions and 3 practice questions. Comprehension questions were derived from the instruction video.

To be eligible for the main experimental task, participants were to answer a designated proportion of questions accurately and had 3 attempts at the full set of 13 comprehension questions. After each attempt at the comprehension questions, feedback was displayed identifying incorrect responses, their respective time stamp for the answer in the video, and a link to the video for review. Each practice question contained an example stimulus with four locations marked with letters. Participants were asked to select which letter contained the most likely hidden location. All incorrect answer selections were followed with corrective feedback. To be eligible for the main experimental task, participants were to answer each of the three practice questions accurately within three tries. Participants who were not eligible to participate were dismissed with partial compensation.

The main experimental task prompted participants to examine 2 to 3 sources of uncertainty in order to find a hidden location. The different uncertainty sources were distinguishable by color: blue and red in the 2-source conditions, plus yellow in the 3-source conditions. Participants were asked to combine the information from all sources and select the most likely location quickly and accurately. This task was developed in JavaScript embedded within the Qualtrics questionnaire and followed a pre-test, practice, post-test design (see Fig. 1). The pre- and post-test included three visualization types, but the practice used just one of the three visualization types.

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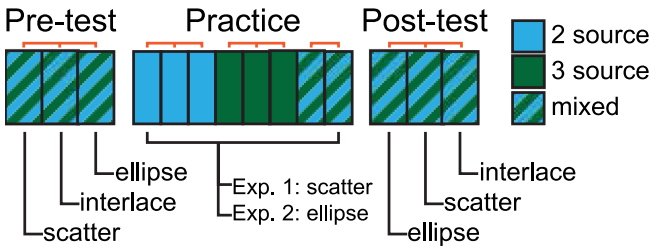


Fig. 1. Diagram of experimental design. Each rectangle represents one block. Brackets above blocks indicate exchangeable blocks under the randomization scheme. Mixed blocks contain both 2 and 3 estimate stimuli in equal numbers.

The pre-test and post-test consisted of 20 trials of each visualization type with 2 sources and 20 trials of each visualization type with 3 sources, for a total of 120 trials per test. Trials were grouped and counterbalanced by visualization type: scatter, ellipse, and interlace. Each trial contained 2 or 3 sources randomly intermixed. The order of stimuli displayed for each trial are randomized within blocks for each participant. For each trial, each stimulus is visible until a response was recorded or timed out (10s) and the next trial was displayed after 0.5s. Pre- and post-testing did not include accuracy feedback.

Practice occurred with a single visualization type and contained 8 blocks of 30 trials each. The first three blocks had 2 sources in each trial, the next three blocks had 3, and the last two blocks were randomly intermixed with 2 or 3 sources. Trial stimuli were displayed until a response was recorded or 10s elapsed. After 0.5s, feedback displayed the participants' selected location, the correct location, a bar showing their cumulative score for that block, and three point scores: accuracy, speed, and total points. Points were generated using clicked response location coordinates and response time. Point feedback was displayed until a participant click was recorded, which then left the location selected, the correct location, and score bar. The next trial began after an additional click. Refer to online supplement [https://osf.io/z8cp4/].

Depending on condition, feedback scores were framed as either point-gains or point-losses. Lastly, participants completed the Intrinsic Motivation Inventory [25] and answered two exit questions regarding enjoyment of the practice task and approach to the task.

2.2 Stimuli

Our test stimuli are abstractions intended to represent scenarios wherein a person must combine estimates of a two-dimensional data item (such as a geographic location on a map) from multiple sources to make a decision. As a real-world example (Fig. 2), consider a search and rescue operation to find a person lost in the wilderness. We might have an estimate of their location (illustrated in blue on a map) based on infrared heat data taken from an overhead drone. We might also have a location estimate (illustrated in red on the map) based on cell phone pings from nearby towers. We might also have a location estimate (illustrated in yellow on the map) based on expert reports of trail hazards. Each of these estimates come with their own uncertainties. How should the search team combine these estimates to find the lost person? A similar scenario could involve determining the most likely location of a hidden explosive, the safest place to withstand a flood, the best spot to set up camp, or the best spot to land a helicopter [26]. The uncertainty integration task would also be the same with any arbitrary type of two-dimensional data (i.e., where the axes represent variables other than cartographic directions). We chose to represent the task in an abstract, scenario-agnostic manner in order to target the basic skills involved in this task and to not restrict implications of our findings to any particular scenario.

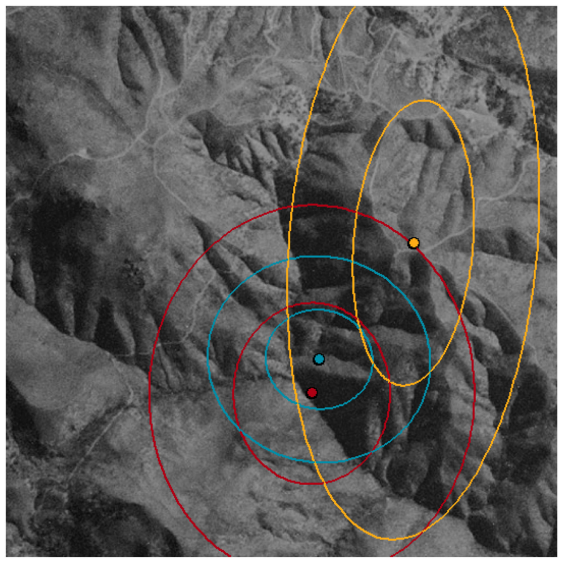


Fig. 2. Hypothetical real-world scenario in which 3 different information sources with different uncertainties must be considered to make a decision. This hypothetical image (not shown to participants) depicts a wilderness area with drone data (blue), cellular data (red), and expert report data (yellow) estimating the location of a lost hiker.

Each stimulus was created by generating two or three independent bivariate distributions and placing them over a two-dimensional space (Fig. 3). For each distribution, the standard deviations in each dimension were drawn from a truncated lognormal with mean parameter -2.0 and standard deviation parameter 0.4 truncated at 0.25. The correlation was drawn from a normal distribution truncated at +/- .8. To position each source, the center of each distribution was shifted by a single random draw from that distribution, and the entire stimulus array was shifted by a draw from a uniform distribution over the central 25% of the stimulus area, subject to the requirement that most (>70%) of the density of each distribution remained onscreen. Stimuli were scaled to 10 cm square and the background was a light gray chosen to have good contrast in all the visualization types.

The visualization types were selected to represent the diversity of approaches to uncertainty visualization reviewed in the Introduction section. Example stimuli shown in the experiments for each type of the visualization presenting two and three uncertainty sources are shown in Fig. 3. Details for the three visualization types follow:

Scatter: We used a static random sample of 600 draws to represent a distributional display. Each sampled point was drawn as a dot. Samples from the two or three sources were distinguished by the color of the drawn dot. Instructions emphasized that dots represent possible outcomes, but that areas with no dots could still be possible outcomes, and that

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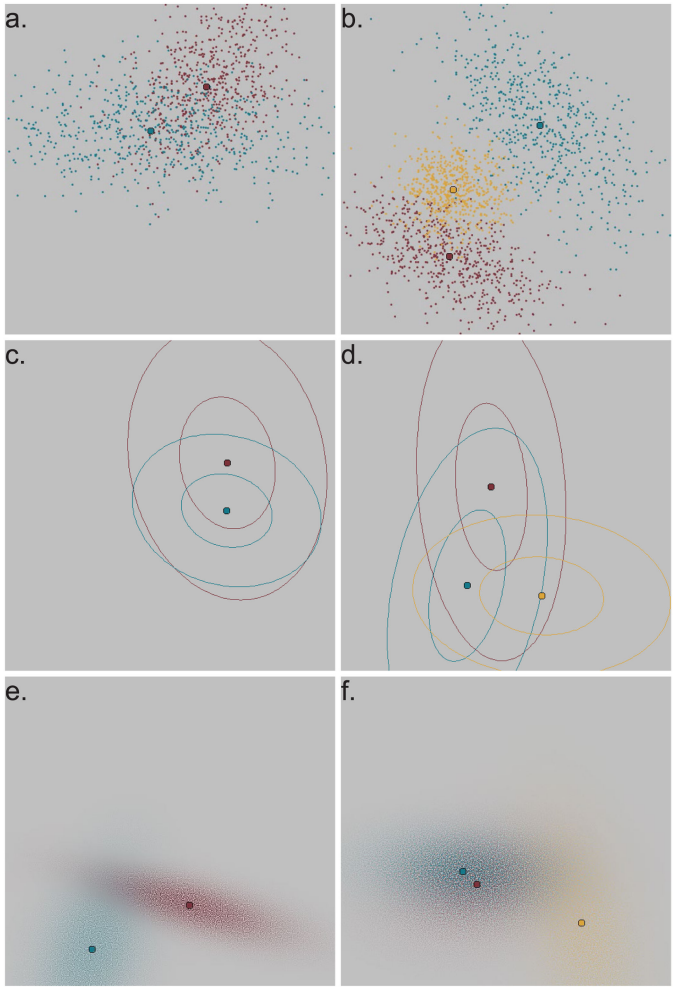


Fig. 3. Example stimuli for each type of visualization. A and B represent the scatter type visualization. C and D represent the ellipse type visualization. E and F represent the interlace type visualization. The visualizations on the left column contain 2 sources while the visualizations on the right column contain 3.

participants should focus on the density of dots in an area rather than the specific locations of individual dots.

Ellipse: The summary display we used in our experiments used concentric ellipses that enclosed the central 50% and 95% of the distributions associated with each individual source.

Interlace: For a distributional way of displaying uncertainty information, we mapped probability density to color value, such that higher probability densities mapped onto brighter, more intense color and the color approached a uniform gray as the probability approached zero. To handle overlapping distributions, we used stochastic block-wise interlacing. Each pixel was assigned to one of the distributions and only displayed the color associated with that distribution. Small (2x2 or 3x3) blocks of pixels were grouped together, such that within each block an equal number of pixels was devoted to each of the distributions in the display.

2.3 Deriving Stimulus Characteristics

Fig. 4 displays how the stimulus difficulty variables are derived from an example stimulus. The example figure

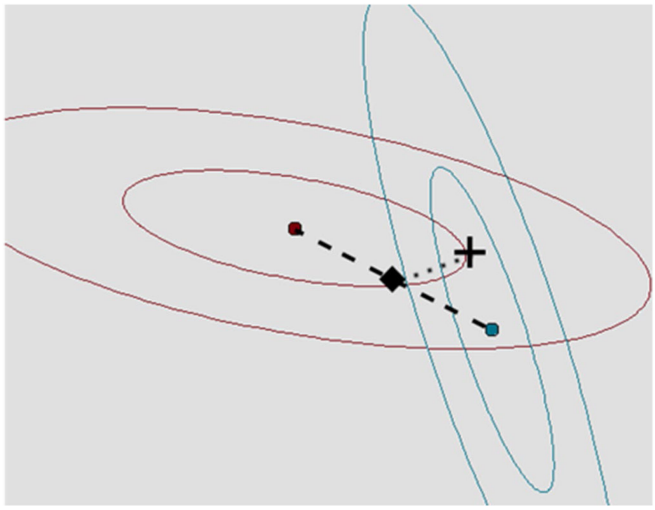


Fig. 4. Derivation of characteristic parameter values from a two-source stimulus displayed in an ellipse type visualization. The plus sign indicates the location of maximum joint density (i.e., the most correct answer), the diamond is the midpoint of all the sources, and colored circles indicate the centers of the red and blue distributions.

shows a stimulus with two sources, one colored red and one colored blue, displayed as an ellipse type visualization. In the center of each source is a dot that indicates its center. We define the measures as follows:

Maximum density at solution: In Fig. 4, the location of the maximum joint density is represented by the plus (+) sign. The value of joint density at the location is used as the parameter.

Distance between source centers: In Fig. 4, it is represented by the length of the dashed line. This parameter is defined as the sum of the Euclidean distance between the midpoint, represented by a diamond (◆) sign, and each source center, represented by the dots of their corresponding color. The location of the source center (represented by dots with colors corresponding to the color of the uncertainty source and a black outline) is obtained by averaging the x and y coordinates of all sources in the stimulus.

Distance between midpoint and solution: In Fig. 4, it is represented by the length of the dotted line. It is the Euclidean distance between the location of the midpoint (◆) and the location of the solution (+)

Since the task involves locating the area with maximum joint density, the variation in the value of maximum density itself could be a factor that contributes to performance score. The measure used for accuracy performance is response ratio, which is the ratio of the joint density at the location the participant clicks on as response and the joint density at the solution. Thus, the penalty for imprecision varies by the relative location and value of joint density at solution.

A simple heuristic that a participant might use when trying to locate the area is to start by anchoring the location to the midpoint between source centers and then adjust according to other information available in the stimulus, such as the relative size, angle, and position of the sources. The amount of information that needs to be processed to adjust this location can be represented by the distance between the midpoint between source centers and the location of the solution. Consequently, the relative distance

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between each of the sources must also be considered. The closer the sources (specifically source centers), the less the participant needs to adjust from their anchor point.

2.4 Experiment 1

2.4.1 Participants

Out of the 189 participants who enrolled in the online experiment via Prolific Academic, 122 participants (77M, 44F, 1Unk) met eligibility criteria and were compensated $25 USD. The mean age was 23.89 years (range 18-60). The median time participants took to complete the survey was 79.11 minutes (range 41.68-180.10 minutes). All testing was approved by the Institutional Review Board of the U.S. Army Research Laboratory protocol number 20-028.

2.4.2 Procedures

In Experiment 1, participants were trained on the scatter visualization type. During practice, participants were given feedback after they clicked on their estimation of the hidden location for each trial. Participants were randomly assigned to feedback framed as either a gain (n=60) or loss (n=62) of points. In the gain condition, feedback was in green text and the score bar increased as points were obtained. In contrast, the loss condition showed feedback in red text and a full score bar that decreased as points were lost with each trial.

2.5 Experiment 2

2.5.1 Participants

Out of 80 participants who were enrolled in the experiment via Prolific, 62 participants (27F, 34M, 1Unk.) met eligibility criteria and were compensated $25 USD. The mean age was 24.98 years (range 18-60). The median time taken to complete the experiment was 86.33 minutes (range 47.88-175.87 minutes). Participants who were enrolled in Experiment 1 were prevented from participating in Experiment 2. All testing was approved by the Institutional Review Board of the U.S. Army Research Laboratory protocol number 20-147.

2.5.2 Procedures. One overall aim for Experiment 2 was to examine effects of focused practice with a different visualization type, ellipses, on uncertainty decision making performance. The methodology and design used for Experiment 2 were the same as Experiment 1 with the exception of a few details. Participants still received 3 tries to pass comprehension, though we relaxed the score needed to pass on the third attempt from 100% to 85% (11/13). This criterion was changed to decrease participant attrition. We added the Cognitive Reflections Test [27] to learn whether performance was influenced by reflective or intuitive strategies, characteristic of using Type 1 and Type 2 processes, respectively.

All participants were assigned to the point-loss condition. In Experiment 1, we did not see effects of framing on performance. Previous work [28] reported an effect of framing in the point loss condition in a similar task. For Experiment 2 we assigned all participants to loss feedback.

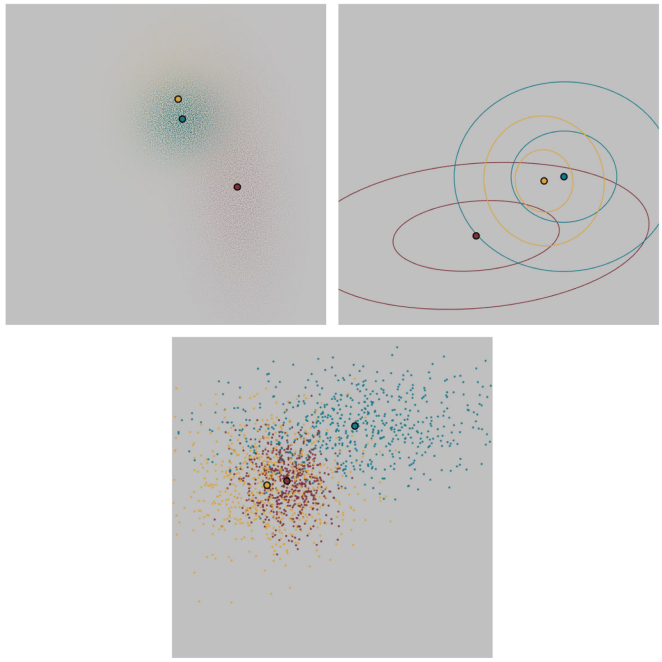


Fig. 5. Three stimuli that differ in type of visualization but share the same maximum joint density, distance between source centers, and distance between the midpoint and solution.

Half of the stimuli in Experiment 2 were retained from Experiment 1. We left some identical stimuli from Experiment 1 to make cross-experiment analyses permissible. The other half were specially modified versions of existing stimuli explained in the next section.

2.5.3 Doppelganger Stimuli. We initially generated the set of stimuli independently across all visualization types for Experiment 1, without consideration of the possible variation due to the random assignment of characteristics. This may result to biased results if one visualization type had more disproportionate amounts of easy or difficult stimuli based on stimulus properties. In order to mitigate these biases, we introduced a new category of stimuli for our second experiment which we refer to as *doppelganger* stimuli.

There are multiple ways to present uncertainty information in our stimuli while still retaining essential characteristics. For example, Fig. 5 displays a set of three source stimuli presenting the exact same uncertainty information. They contain the same bivariate normal distributions with the same relative position to each other and consequently the same location of maximum density (i.e., the solution of each stimulus). In addition to varying the visualization type of each stimuli, we can also transform the orientation of each image by rotating, flipping, or a combination of both. In order to retain how much density is within the image, we limit the stimulus rotation to be either a 90, 180, or 270 degree rotation. In total, we devised eight unique ways to transform the orientation of a stimulus. Additionally, we also changed the configuration of source colors for some stimuli. We call these doppelganger stimuli because they represent identical information and have identical values

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for the three difficulty variables, but they are otherwise transformed to mask their similarities.

In Experiment 2, we devised a set of twenty original stimuli (even split between two and three source stimuli) and transformed them across the three visualization types such that each original stimulus had a doppelganger across all visualization types in both pre- and post-block. This means that each original doppelganger stimulus has six different transformed versions across the experiment. The remaining stimuli were identical to those used in Experiment 1; we refer to that subset of stimuli as *twins*.

2.6 Analysis

The performance measure for accuracy that is used as a dependent variable in the analysis is response ratio. Response ratio is the ratio of the probability density at participants' clicked location as response in a stimulus to the density at the solution, i.e., the maximum density for the particular stimulus. The response time (the time elapsed between when a stimulus is displayed and when a participant clicks on it) is also used as a dependent variable. On trials in which the 10s maximum time was reached, the response was scored as a response ratio of zero and a missing response time.

For response ratio analyses, we developed a Bayesian beta regression model using the *Stan* language and the *cmdstanpy* interface [29]. Response ratio was modeled with a beta function using the mean and precision parameterization. The mean was formulated as a linear combination of predictors and subject to an inverse logit transformation. The predictors were a block-specific intercept, with blocks defined by the visualization type, (scatter, ellipse, or interlace), block-type (pre- or post-practice block), and number of uncertainty sources (2 or 3) for a total of 3x2x2=12 independent intercept estimates with standard normal priors. The three measures of stimulus difficulty (maximum density, distance of solution from midpoint, and distance between source centers) and the individual differences measures (regulatory focus prevention and promotion, graph literacy, numeracy, the interaction of graph literacy with numeracy, and in Experiment 2 only, cognitive reflections task scores) were z-transformed and added as independent variables with standard normal priors on the coefficients. A random effect of stimulus was included, with mean zero and half-normal prior on the standard deviation. Potentially correlated random effects of subject by block-type interaction were included, with half-normal priors on the standard deviations and a uniform prior on the correlation coefficient.

For response time analysis, a similar approach was taken using a truncated lognormal model for response time, such that the log of response time was modeled as normal with a mean parameter and a standard deviation parameter. As with response ratio, the mean parameter in the RT model was a linear combination of a block-specific intercept, stimulus difficulty, individual differences measures, and random effects of stimulus and participant. The priors were identical with the exception that the block-specific intercepts had normal (0.5, 1) priors (implying a modal response time of 1.65s). The standard deviation parameter was formulated as a linear combination of a block-type intercept (pre-practice or post-practice) with normal (1, 0.5) priors and a potentially correlated random effect of subject by block-type interaction with a normal (0, 0.5) prior on the standard deviation and a uniform prior on the correlation, and the standard deviation parameter was subject to an exponential transformation.

For each model, samples were generated using the No-U-Turn Sampler [30] with four independent chains with 1000 warm-up and 1000 post-warm-up iterations. Sampling completed in all cases with no divergences, and split R-hat was less than 1.1 for all parameters. To identify outliers, we computed average logit-transformed response ratio for each participant with each visualization type on the pre-test and the post-test and applied an adjusted boxplot rule [31] with a fence set at 2.0 to detect very low performance. Any participant identified as having very low performance on any block was excluded from analysis entirely.

3 Results

Models were fit for speed and accuracy using five different sets of trials: Experiment 1 all trials, Experiment 2 all trails, Experiment 2 doppelganger trials, Experiment 1 twin trials and Experiment 2 twin trials. Response accuracy and speed for pre-practice blocks and post-practice blocks are presented for Experiments 1 and 2, followed by a comparison of performance on twin stimuli in both experiments. Details of random effects and precision parameters are available in an online supplement [https://osf.io/23af9/], and they will not be discussed here. Analysis on individual difference measures collected from this study did not yield any notable findings and can also be viewed in the online supplement [https://osf.io/y2pm6/]. Finally, the effects of participant characteristics and stimulus characteristics on performance are summarized.

3.1 Experiment 1: Accuracy and Speed

Experiment 1 included 122 participants who completed the task, of whom 4 were identified as outliers and excluded from analysis.

Experiment 1 included a between-subjects manipulation of the framing of feedback delivered after each practice trial as point gains or point losses. We found this manipulation to have no clear effect on speed or accuracy of performance, so it will not be discussed further here, but detailed analysis of the effects of framing are available online at [https://osf.io/tukpx/].

3.1.1 Descriptive

Individual participant accuracy was summarized with the average logit-transformed response ratio. Response time was summarized with a 20% trimmed mean. Accuracy and response time for the three visualization types both before and after practice with the scatter stimuli appear in Fig. 6.

3.1.2 Modeling

The beta regression of response ratio included within the average response ratio term intercept parameters for each combination of visualization type (ellipse, interlace, scatter),

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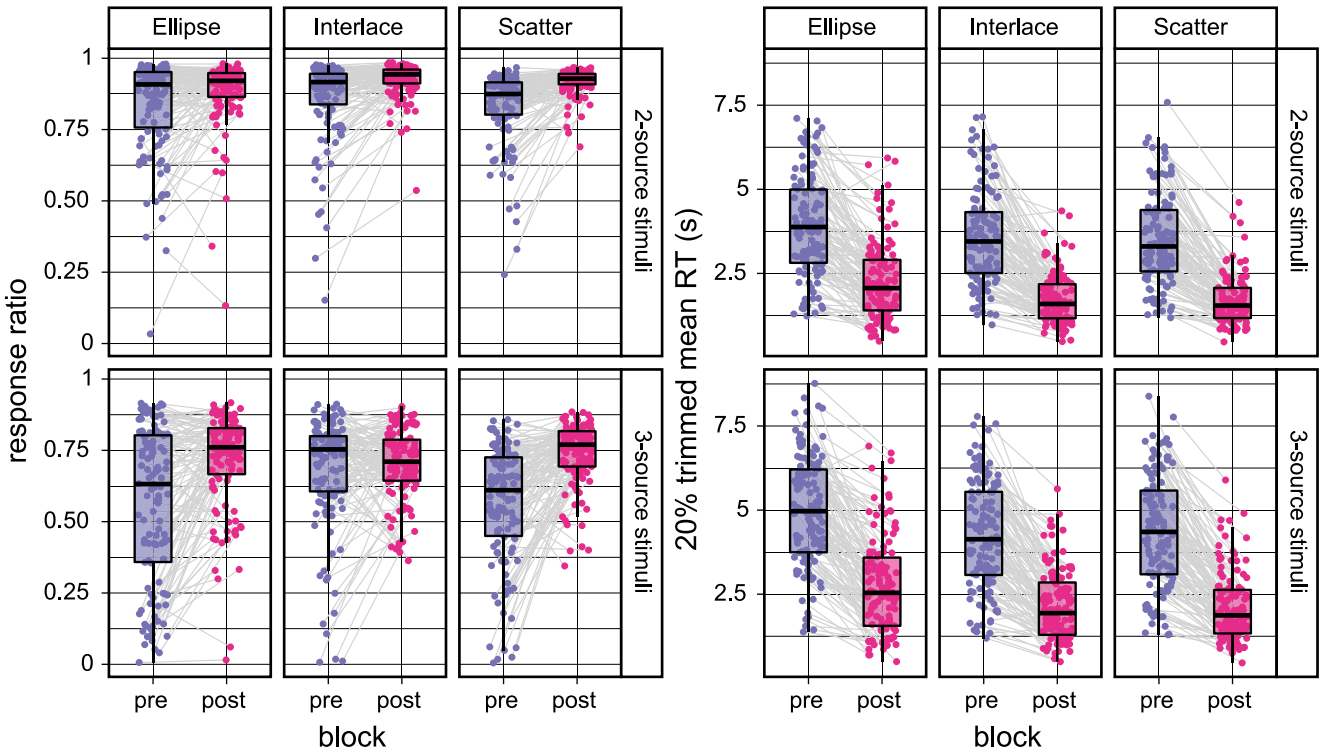


Fig. 6. Summaries of response ratio and response times in Experiment 1. Dots (N=118) show the inverse logit of the mean of logit-transformed response ratios and 20% trimmed mean response time on the left and right panels, respectively. Boxes show group median and 25th and 75th percentiles, with whiskers showing extrema excluding points more than 1.5 times the inter-quartile range from the extents of the boxes. Summaries are separated by visualization type, number of sources, and before or after practice.

number of sources in the stimulus (2 or 3), and whether the block was before or after the practice session (pre- or post-). The same intercept terms were included in the truncated lognormal regression of response time within the term describing the mean of the underlying normal distribution. These parameter estimates are shown in Fig. 7. Post-practice performance was more accurate and faster following practice for all visualization types and both numbers of estimates. Responses were faster and more accurate for two-source stimuli compared to three-source stimuli in all cases. Post-practice accuracy was similar for all visualizations with two-source stimuli. Post-practice accuracy was somewhat lower with interlace stimuli (posterior mean .71, 95% credible interval [.67, .74]) compared to ellipse (.73, [.70, .76]) and scatter (.74, [.71, .77]) in the three-source stimuli. In both pre- and post-practice blocks, responses to ellipse stimuli were slower than responses to the other two types.

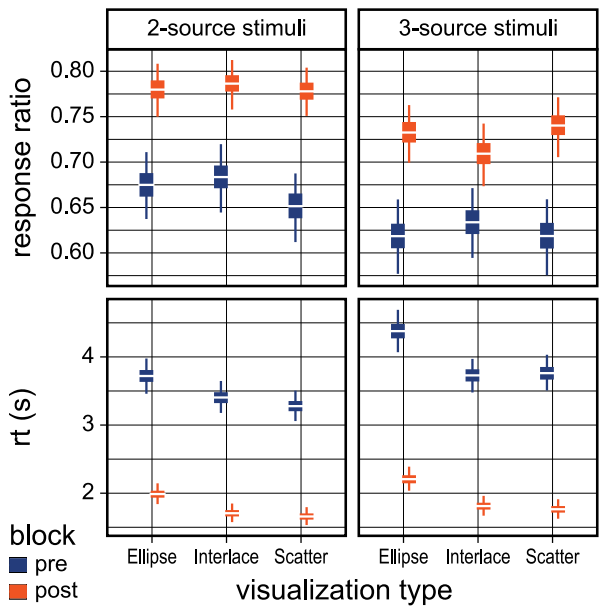


Fig. 7. Posterior summaries for effects of visualization type, number of sources, and pre- or post-practice blocks on response ratio and response time. Boxes and whiskers show a central 50% and 95% credible interval, respectively, and the white central mark shows the mean of the posterior distribution. The upper panels show coefficients for response ratio. The lower panels show coefficients for response time in seconds.

3.2 Experiment 2: Accuracy and Speed

Experiment 2 included 62 participants who completed the task, of whom 6 were identified as outliers and excluded from analysis.

3.2.1 *Descriptive*

Individual participant accuracy was summarized with the average logit-transformed response ratio. Response time was summarized with a 20% trimmed mean. Accuracy and response time for the three visualization types both before and after practice with the ellipse stimuli appear in Fig. 8.

3.2.2 *Modeling*

The modeling approach for Experiment 2 was identical to Experiment 1, except two separate analyses were carried out: one using data from all stimuli, and the other only

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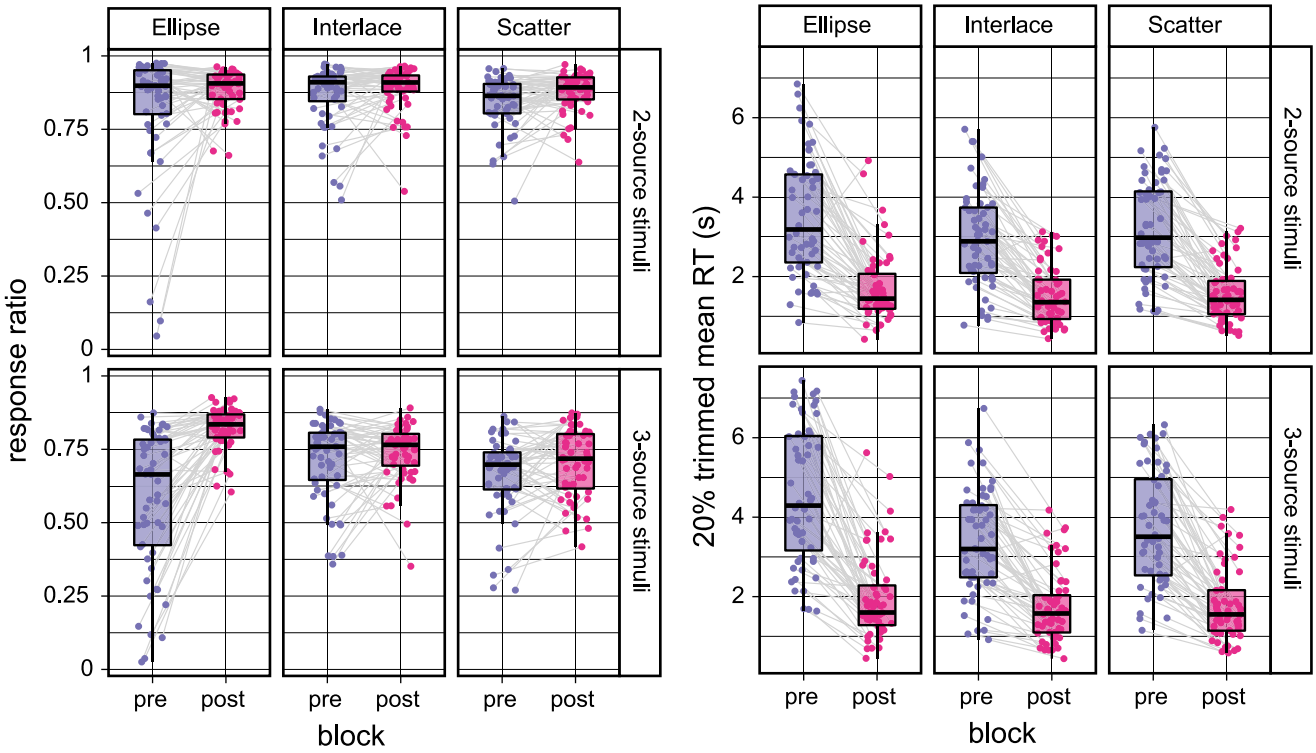


Fig. 8. Summaries of response ratio and response times in Experiment 2. Dots (N=56) show the inverse logit of the mean of logit-transformed response ratios and 20% trimmed mean response time on the left and right panels, respectively. Boxes show group median and 25th and 75th percentiles, with whiskers showing extrema excluding points more than 1.5 times the inter-quartile range from the box. Summaries are separated by visualization type, number of sources, and before or after practice.

using data from doppelganger stimuli. The analysis of doppelganger stimuli was intended to eliminate variability of stimulus difficulty as a possible contributor to differences between visualizations and pre-/post-practice blocks. Results of accuracy and response time models for Experiment 2 appear in Fig. 9. In what follows, we report the parameter estimates associated with the full stimulus set except where they differ from the doppelganger estimates, in which case both are reported. Post-practice speed was faster compared to pre-practice speed with all visualizations and number of sources. In post-practice blocks, speed with ellipse stimuli was slightly lower than with the other two visualizations.

Post-practice accuracy was substantially higher than pre-practice accuracy for all visualizations in the 2-source stimuli. With 3-source stimuli, post-practice performance with ellipse stimuli (posterior mean .79, 95% credible interval [.75, .83]) was much higher than pre-practice with ellipse (mean .63, [.59, .68]) and also was higher than post-practice accuracy with interlace (mean .72, [.69, .76]) and scatter (mean .72, [.68, .75]). Pre-practice accuracy with interlace and scatter 3-source stimuli (mean .68, [.64, .72] and mean.68 [.64, .72], respectively) was relatively high, and was not substantially lower than respective post-practice accuracy. Within the doppelganger stimulus set, pre-practice accuracy was even higher and more similar to post-practice accuracy for 3-source interlace (mean .71, [0.66, .76] pre; mean .74 [.70, .78] post) and Scatter (mean 0.70 [0.65, .75] pre; mean 0.71 [.66, .76] post) stimuli.

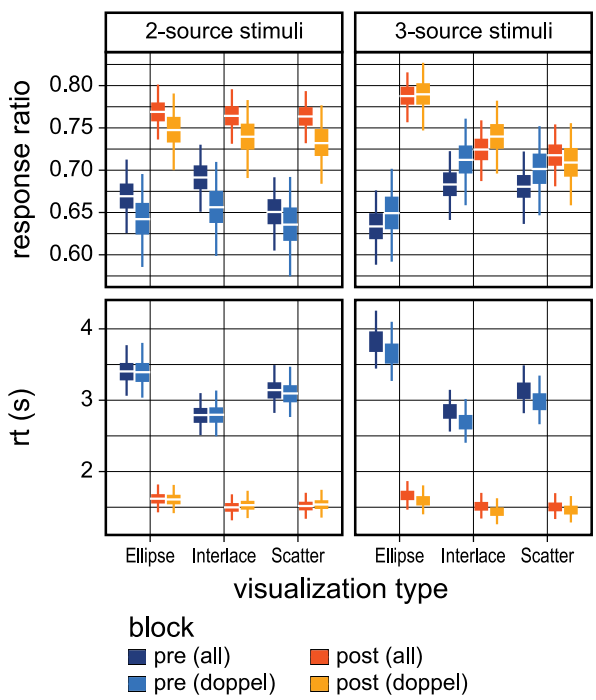


Fig. 9. Effects of visualization type, number of sources, and pre- vs. post-practice on accuracy and speed of responses. Results are from analysis of the full dataset as well as an analysis only of doppelganger stimuli. Boxes and whiskers show a central 50% and 95% credible interval, respectively, and the white central mark shows the median of the posterior distribution. The upper panels show coefficients for response ratio. The lower panels show coefficients for response time in seconds.

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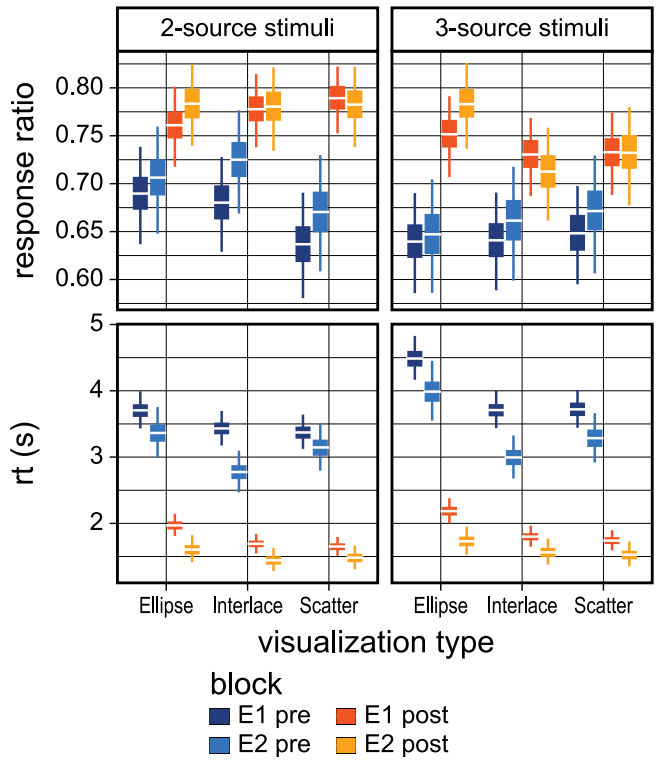


Fig. 10. Posterior estimates using data from stimuli appearing in both experiments. Colors indicate blocks before (pre) and after (post) the practice session as well as Experiment 1 (E1) in which participants practiced with the scatter visualization or Experiment 2 (E2) in which participants practiced with the ellipse visualization. Boxes and whiskers show a central 50% and 95% credible interval, respectively, and the white central mark shows the median of the posterior distribution. The upper panels show coefficients for response ratio. The lower panels show coefficients for response time.

3.3 Comparing Experiments 1 & 2 (twins)

To facilitate a fair direct comparison between Experiments 1 and 2, half the stimuli in Experiment 2 were identical to those in Experiment 1. Models were independently fit on this subset of stimuli for both Experiment 1 and 2. These models had the same structure as those already described. Posterior distributions of the intercept terms for the average performance term in the response ratio models and average parameter in the lognormal response time models appear in Fig. 10.

Consistent with the full dataset analyses, responses were faster and more accurate after practice across the board. Experiment 2 tend to show faster and more accurate on the pre-practice blocks than responses in Experiment 1 to various degrees. Instructions, stimuli, and context were identical for this comparison. In the post practice-block, the Experiment 2 responses were faster than those in Experiment 1, although in both cases responses to ellipse stimuli were slower than to the other two visualizations.

In the two-source stimuli, post-practice accuracy was similar across the two experiments, although practice with scatter stimuli in Experiment 1 was associated with slightly worse post-practice performance with ellipse stimuli (posterior mean .76, 95% CI [.72, .80]) compared to Experiment 2 (.78, [.74, .82]). In the three-source stimuli, post-practice accuracy was somewhat higher for ellipse stimuli in Experiment 2 (.78, [.74, .83]) compared to Experiment 1 (.75, [.71, .79]), and post-practice accuracy was slightly lower for interlace stimuli in Experiment 2 (.71, [.66, .76]) compared to Experiment 1 (.73, [.69, .77]). Post-practice accuracy on three-source scatter stimuli was equivalent between Experiment 1 (.73, [.69, .77]) and Experiment 2 (.73, [.68, .78]).

TABLE 1
Stimulus Characteristics

	Experiment 1		Experiment 2	
	m	*SD*	*M*	*SD*
Max. density	35.687	15.565	36.590	13.882
Source distance	0.303	0.169	0.290	0.147
Mid. distance	0.047	0.031	0.048	0.029

3.4 Stimulus Characteristics

Stimulus characteristics were included in both the accuracy and the response time models to understand how stimulus properties related to performance. The stimulus properties included were the density of the joint distribution over all sources at the optimal location, the Euclidian distance between the optimal location and the midpoint between all source centers, and the average distance between source centers. These properties were standardized; means and standard deviations of these property measures appear in Table 1. Summaries of the posterior distributions for stimulus characteristics fit to the full datasets for Experiments 1 and 2 appear in Fig. 11.

Although there were some differences in coefficient estimates between Experiment 1 and Experiment 2, both showed similar overall patterns of results. Larger values of all stimulus characteristics were associated with less accurate and slower performance overall, with pairwise source distance having the largest coefficient on both speed and accuracy. The distance of the solution from the midpoint had the smallest effect on both speed and accuracy.

4 Discussion

In two experiments we found that a single short practice session with feedback substantially increased response accuracy and reduced response time on a two-dimensional spatial uncertainty integration task. Experiments 1 and 2 used some different stimuli, but analyses of data only from stimuli appearing in both experiments showed that post-practice performance was nearly equivalent in the two experiments, with the exception that those who practiced with the ellipse visualization performed somewhat better on 3-source ellipse trials and to a lesser extent worse on the 3-source interlace trials compared to those who practiced with the scatter visualization.

A two-stage theory of decision-making with visualized data [10] distinguishes between extracting information from a display and making an inference with that information. The observation that performance increases transferred to non-practiced visualizations supports the interpretation that participants were primarily improving on the inference task. If the practice were just improving the extraction of information from the visualization, then we would not have expected performance gains to transfer to non-practiced

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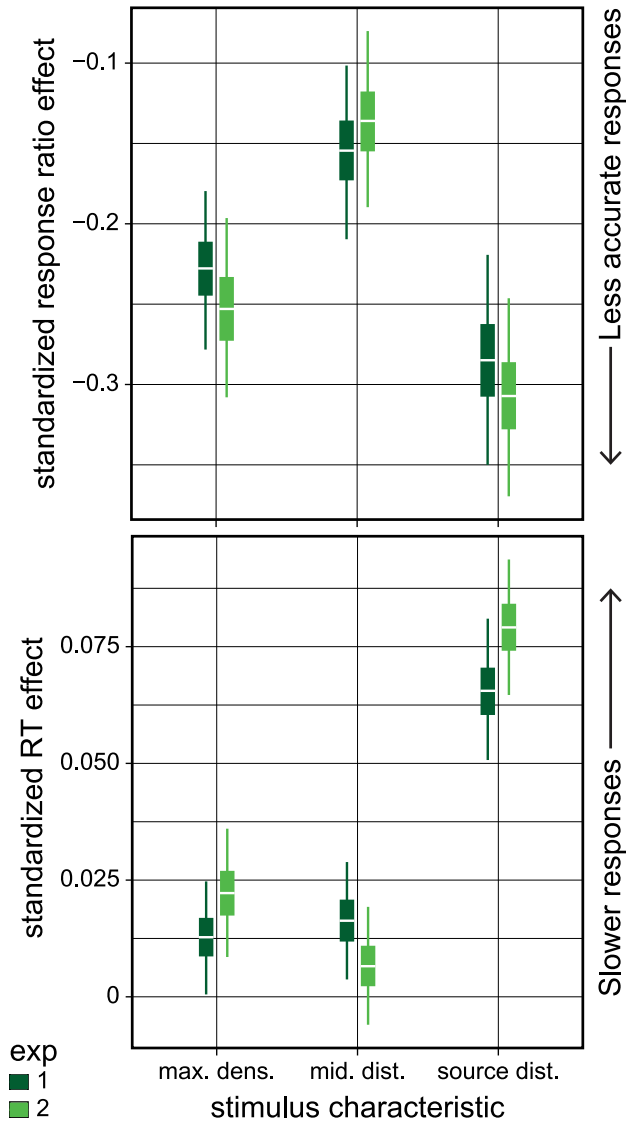


Fig. 11. Effects of stimulus characteristics on response ratio and response time. Stimulus characteristics are the density of the joint distribution at the optimum (max. dens.), distance of the solution from the simple average of source centers (mid. dist.), and average distance between source centers (source dist.) Boxes and whiskers show the central 50% and 95% of the posterior distribution. The white central mark shows the mean of the posterior distribution.

visualizations. However, practicing with the ellipse stimuli resulted in a performance advantage with 3-source ellipse stimuli in the post-test at the apparent cost of a disadvantage with 3-source interlace stimuli. Summary data displays, such as confidence intervals and the ellipse stimuli we used, are often misunderstood [32], [33]. They are also unintuitive, in that size is conflated with uncertainty, violating the salience principal [34], [35], [36]. However, the ellipse stimuli here also have the advantage of being concise, in that they convey a lot of information with minimal 'ink' [37]. It is plausible that iterated practice with the ellipse stimuli allowed participants to overcome those difficulties to realize the benefits of a more concise visualization.

We also examined characteristics of individual stimuli to see if they had systematic effects on performance. We found that stimuli with smaller distances between source centers were associated with more accurate and faster performance and that stimuli with a smaller maximum posterior density were associated with more accurate but not faster performance. The effect of source distances on performance suggest that the task is more difficult when the stimulus centers are farther from each other, which requires making inferences farther out in the tails of the represented distributions. The association of larger posterior density at the optimal solution with worse performance could reflect that a lower, broader peak in the posterior density is somewhat more forgiving of imprecise responding, but it might also partially counteract the effects of source distances, as all other things being equal, farther apart sources should have lower maximum densities.

4.1 Implications

The results of our experiments established evidence that processing uncertainty information can be enhanced by practice with feedback. Furthermore, the fact that the increase in performance transfers to other types of visualization adds even more utility towards providing such kinds of training.

While the task we chose was deliberately abstract, spatial uncertainty is fundamental in a wide range of real-world contexts such as weather forecasting, route planning, reconnaissance, city planning, battlefield strategy, natural resources exploration, farming, climate modeling, and ecology. For example, a homeowner needs to understand the uncertainty of a hurricane path in order to decide whether to evacuate [38]. A city planner needs to understand uncertainties in likely flooding areas in order to guide construction and safety planning [39]. An ecologist needs to understand uncertainties in habitat quality and species ranges to know which areas are most critical for conservation [40], [41]. A military commander needs to understand spatial uncertainty in the locations of hazards and adversaries in order to plan a mission [42].

Complex situations require more complex information, which makes understanding and integrating uncertainty a valuable skill. Our findings suggest that this skill can be significantly improved with brief practice, which makes improved uncertainty integration and decision making within reach of practitioners across different fields. For example, health care companies could provide uncertainty integration training for staff to help them make medical decisions and to help explain options to their patients. Schools could include this training in their curricula to enhance these skills in different contexts. The military could include such training as part of employee development for different job tracks.

Our study focused on spatial uncertainty, but many forms of non-spatial uncertainty are also critical. These can include medical (e.g., a drug's effectiveness), financial (e.g., stock market predictions), political (e.g., election outcomes), safety (e.g., environmental sensor readings), class uncertainty (e.g., threat vs. non-threat), and others. In a mathematical sense, multi-dimensional uncertainty can be visualized spatially, so practicing tasks on spatial uncertainty seems like a promising place to start in order to explore the role of training in reasoning with uncertain information.

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4.2 Limitations and Future Research

This experiment was online, so we did not have control over the hardware used to display the stimuli or over the environment in which the experiment took place. Differences in color calibration could have particularly affected the interlace stimuli which used color differences to convey likelihood. As such, results with that stimulus type might be regarded with some caution. Practical applications of similar visualizations might face similar challenges, however.

We observed unexpected differences in pre-practice performance between Experiment 1 and Experiment 2. Our analyses of stimuli in common suggests that some of this can be accounted for by the different stimuli in the two experiments, but some pre-test differences remained. Although both experiments used the same recruiting approach, there were potentially relevant differences. Recruitment took place at different times (Summer of 2020 for Experiment 1 and Fall 2020 for Experiment 2), so the participant pool was potentially different[1]. An additional difference between Experiment 1 and 2 is that we lowered the passing criterion for the comprehension check in Experiment 2; however, the lower criterion was applied to the group with better pre-practice performance, so this leads to no easy explanation.

Our stimuli were sampled from a fairly high dimensional space, with 10 and 15 parameters describing the source configurations in 2- and 3-source stimuli, respectively. In a limited number of trials, we cannot but sparsely sample this parameter space. We attempted to control for differences in difficulty among the different stimuli with the derived stimulus properties of the distance between sources, the distance from the midpoint among the sources to the best solution, and the posterior density of the solution. Although these predictors were related to performance on those stimuli, considerable variability was still observed. The derived measures we used were somewhat ad hoc, and other derived measures could be computed.

An additional limitation is that the uncertainties always followed bivariate Gaussian distributions. Future work will need to examine whether our findings generalize to other kinds of distribution.

In the present context, the location estimates and their uncertainty capture the only information relevant to the decision. As such, an exactly correct solution can be computed with a simple formula, which might bring into question why a human might need to be involved at all. In practice, we expect that a human decision-maker might have additional information, assumptions, or beliefs that might not be easily incorporated into such a computation. In that more practical scenario, the uncertainty integration skills examined here could be critical to arriving at an optimal solution, although this hypothesis remains to be tested.

5. Conclusion

Here we demonstrated in two experiments that a brief practice session with feedback led to better performance on an uncertainty integration task. Practice with an ensemble visualization resulted in similar post-practice improvements on the practiced and unpracticed visualizations. Practice with a summary visualization resulted in superior post-practice performance using the summary visualization compared to the non-practiced visualizations as well as the performance achieved by those who practiced with an ensemble visualization. These results suggest that practice improved participants' skill with the uncertainty integration task, rather than only improving their use of a specific visualization approach. We see these results as pointing toward a practical recommendation to help people make better decisions with visualized uncertainty: provide easy opportunities for iterated practice with feedback. This approach will not always be possible, but where it is, it could lead to faster, beter decisions under uncertainty.

1. There was also the global COVID-19 pandemic during that time, so we shall not attempt to generalize to all Summers and Falls.

Acknowledgments

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